

GAYATHRI D

+91 7598092785 | gayatrisenthamarai@gmail.com

[GitHub](#) | [LinkedIn](#)

RESEARCH SUMMARY

Computer Science undergraduate pursuing the Quantum Computing Vertical (Minor), actively researching Quantum Machine Learning (QML) with a focus on noise-aware feature selection for Quantum Convolutional Neural Networks (QCNN). Experienced in Qiskit-based circuit design, PCA-driven dimensionality reduction, and quantum noise-channel modeling (bit-flip, depolarizing, phase-flip) for NISQ-era hardware. Solid foundation in linear algebra & probability for quantum computing, backed by a strong software engineering background across full-stack development.

EDUCATION

Rajalakshmi Institute of Technology (RIT), Kuthambakkam, Chennai **Expected 2028**
B.E. Computer Science and Engineering | Quantum Computing Vertical (Minor)

Cumulative GPA: 8.37 / 10

Core Foundations: Quantum Circuit Fundamentals, Linear Algebra & Probability for Quantum Computing, PCA-based Dimensionality Reduction, Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks.

Higher Secondary Certificate — 91.5% **(2022 - 2024)**

Bharathiyar Matric Higher Secondary School, Attur

Secondary School Leaving Certificate — 88.6% **(2020 - 2021)**

Bharathiyar Matric Higher Secondary School, Attur

TECHNICAL SKILLS

- Quantum & Math: Qiskit (circuit design, simulation, execution), Quantum Machine Learning, QCNN architectures, noise-aware feature selection, PCA, RY rotational encoding, noise modeling (bit-flip, depolarizing, phase-flip), Linear Algebra & Probability
- Languages: Java, Python, JavaScript, SQL, C (Fundamentals)
- Frameworks & Toolkits: Spring Boot, FastAPI, Flask, Node.js, Express.js, React, Next.js
- Systems & Tools: Git, GitHub, Docker, Maven, Postman, Vercel, Render, MySQL, PostgreSQL, MongoDB

QUANTUM & ADVANCED RESEARCH PROJECTS

Noise-Aware Feature Selection for Quantum Convolutional Neural Networks (NAFS-QCNN)

Research Project | Quantum Machine Learning

Paper Communicated

- Proposed and experimentally evaluated a Noise-Aware Feature Selection (NAFS) framework for QCNNs, using the scikit-learn handwritten-digit dataset (binary classification, 360 samples).
- Applied PCA to reduce inputs to 8 quantum features, implemented RY rotational encoding, and designed a hierarchical $8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ QCNN architecture (28 trainable parameters).
- Ranked features by behavior under bit-flip, depolarizing, and phase-flip noise; selected 6 noise-aware features, achieving a 25% reduction in feature representation.
- Validated via an ablation study (Standard QCNN vs. NAFS-QCNN vs. Random-6 QCNN) using classification accuracy and a 95% reliability boundary across noise strengths 0.00–0.30.

OTHER PROJECTS

Enterprise Learning Platform

2024

- Developed a full-stack learning and career guidance platform with courses, assessments, coding practice, certifications, and career roadmaps using React, Spring Boot, MySQL, JWT, and Google Gemini API.

Job Portal

- Built a full-stack job portal enabling registration, resume upload, and job applications, with role-based access and backend integration.

EXPERIENCE

Zidio Development — Java Full Stack Developer

2026

- Built REST APIs, implemented authentication mechanisms, and integrated frontend components with backend systems using Java, Spring Boot, and MySQL.

Infosys Springboard Virtual Internship — Java Full Stack Developer

- Completed a virtual internship focused on full-stack web development using Java, Spring Boot, and MySQL.
- Developed RESTful APIs with database integration and connected them to a responsive frontend.
- Applied core software engineering practices, including version control with Git and GitHub and API testing with Postman.

PUBLICATIONS

- “Noise-Aware Feature Selection for Quantum Convolutional Neural Networks (NAFS-QCNN)” — Gayathri D. Communicated for publication.
- “Intelligent Risk Assessment Model for Accident Prevention in Hazardous Industries Using Machine Learning” — Dharshini V, Dhathri M, Gayathri D, Gobikaa E, and Bharathy S. Scopus Indexed Publication, 2026.
- IEEE Conference Paper: “Risk Analysis in Chemical Industries” — presented at an IEEE conference.

ACHIEVEMENTS & LEADERSHIP

- Solved 200+ LeetCode and 1500+ SkillRack problems; completed the 100 Days LeetCode Challenge with documented solutions and complexity analysis.
- Maintain 20+ GitHub repositories spanning backend, full-stack, and quantum ML projects.
- NPTEL Elite certifications in Design & Analysis of Algorithms and Machine Learning, plus DSA using Python.
- Completed Microsoft Foundations of AI Internship; certified in UiPath, n8n Workflow Automation, Celonis, and MongoDB (Basics, Relational to Document Model).
- Event Organizer & Coordinator for college technical events; Club Member, Infinitus Club (Math), Dept. of CSE.